Vidyavardhini's College of Engineering & Technology
Department of Computer Science & Engineering (Data Science)

MDC501 Wireless and Mobile Communication

Module 1

Wireless Transmission Principles, Frequency Spectrum Allocation and Introduction to Cellular Communication

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Lecture Objectives

After this lecture, students will be able to:

- Explain the need for wireless communication
- Describe the elements of a communication system
- Understand the basic concept of wireless transmission
- Explain path loss (qualitative understanding)
- Identify ITU frequency bands and licensed/unlicensed allocation
- Understand the cellular concept and why hexagonal cells are used
- Explain frequency reuse, cluster size and the reuse ratio
- Understand cell splitting and sectoring as capacity techniques

Section 1.1

Wireless Transmission Principles

Need for Wireless Communication

- **Definition:** Wireless communication is the transfer of information between two or more points without a physical (wired) connection, using electromagnetic waves.

Why is it needed?

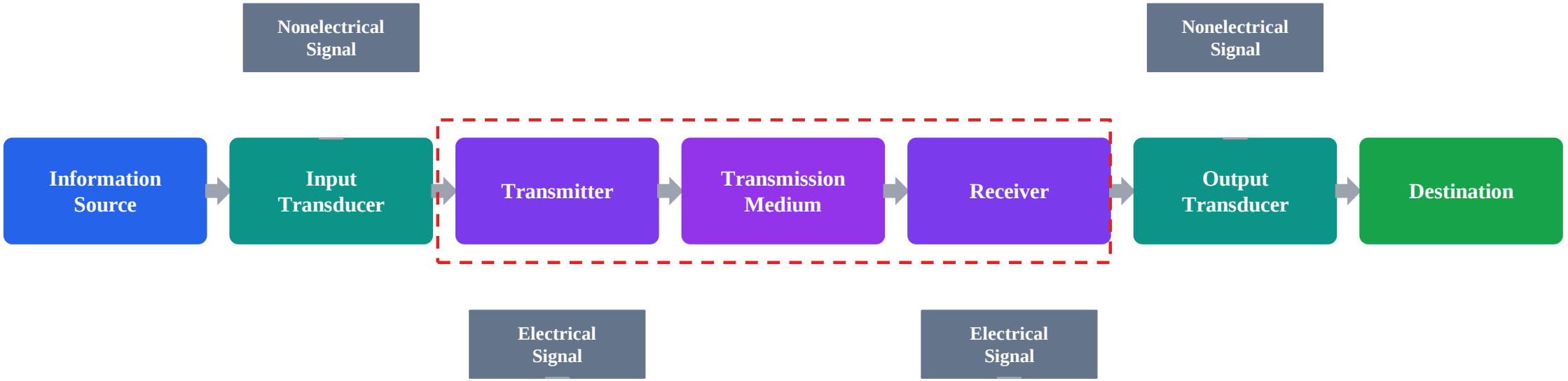
- Mobility – access anytime, anywhere
- Fast deployment and easy installation
- Long-distance communication
- Supports IoT, smart devices and automation
- Critical for defense and emergency communication

Real-world applications:

Mobile communication • IoT devices • Defense systems • Smart homes

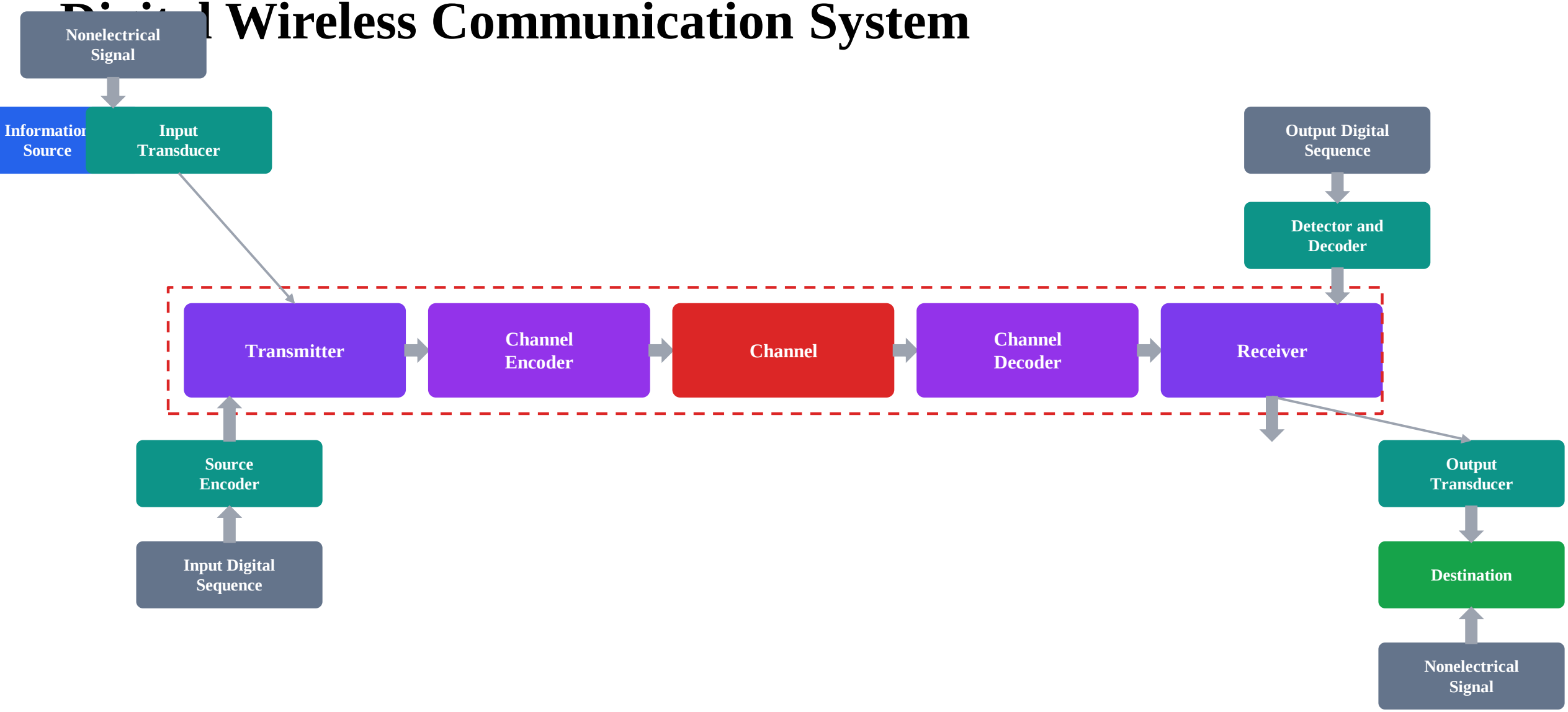
Elements of a Communication System

- **Definition:** A communication system is an arrangement of individual elements that together enable the transfer of information from a source to a destination.



- Transmitter performs 3 functions: modulation, encoding and transmission of the signal.
- Transmission medium examples: wire, coaxial cable, optical fiber, radio link.
- Receiver performs 3 functions: reception, decoding and demodulation (reverses the Tx process).

Digital Wireless Communication System



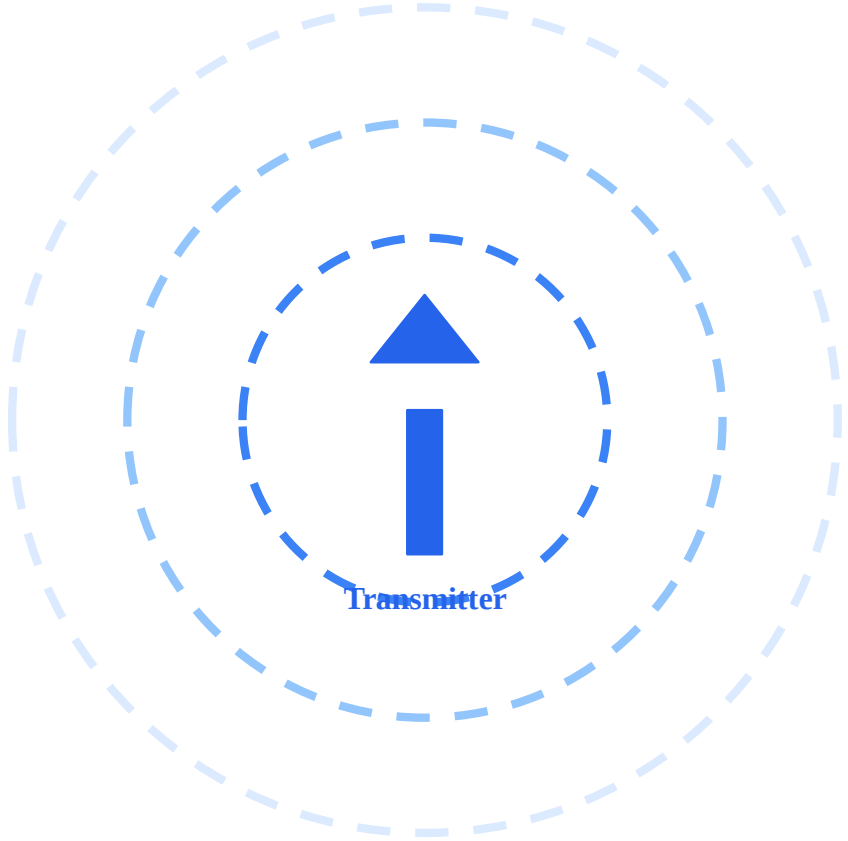
Channel Encoder adds error-control coding for reliable transmission; Channel Decoder and Detector reverse this at the receiver to recover the original digital sequence.

Wireless Transmission & Path Loss

- Wireless transmission uses electromagnetic waves to carry signals through air — no physical cable is required.

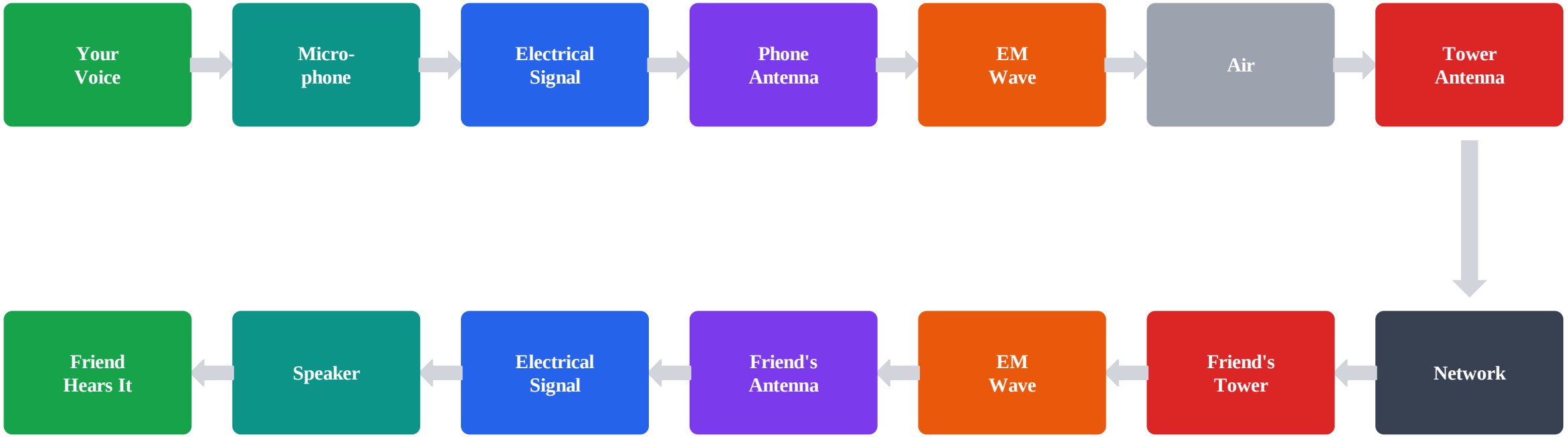
Path Loss (qualitative):

- Reduction in signal strength as it travels away from the source
- In free space, power falls off gradually with distance
- In a real wireless channel, obstacles (buildings, trees, terrain) make the signal weaken faster than in free space
- Example: Wi-Fi signal gets weaker as you move away from the router



Signal strength decreases with distance

How Your Voice Reaches Your Friend



Sound is converted to an electrical signal, then to an electromagnetic wave that travels through the air to a tower, across the network, and is converted back to sound at the other end.

Section 1.2

Frequency Spectrum Allocation

Electromagnetic Spectrum: ITU Band Classification

- **Definition:** the electromagnetic spectrum is the complete range of frequencies of electromagnetic radiation, from very low frequencies to gamma rays.

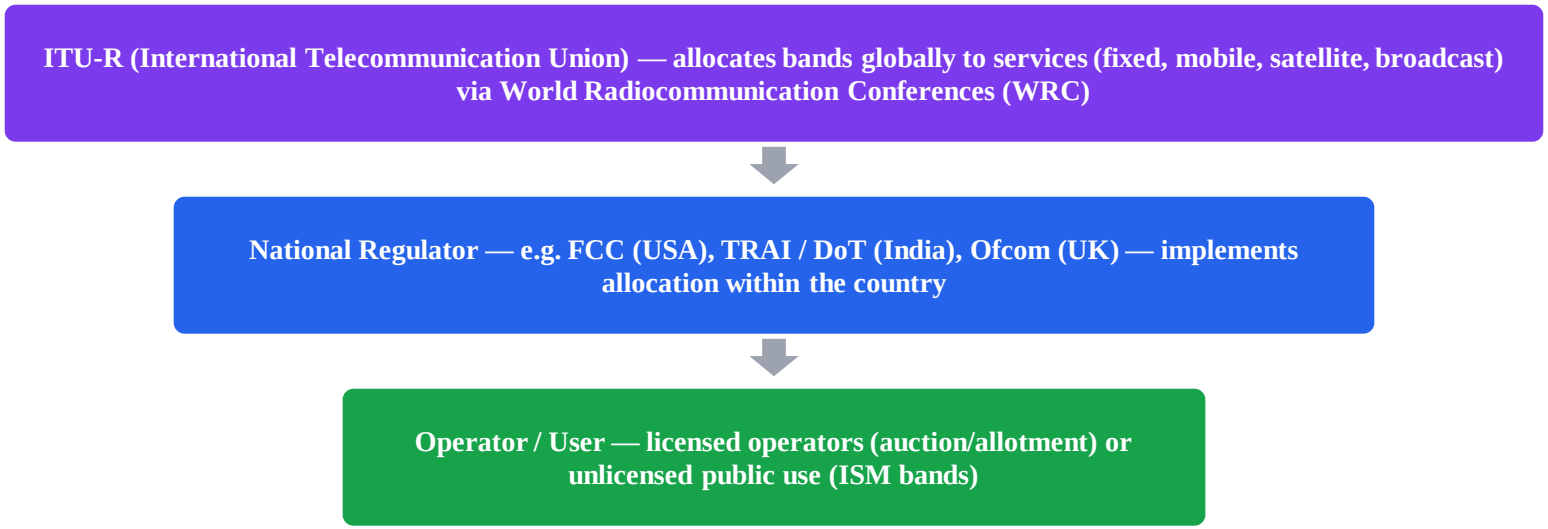
Band	Frequency Range	Typical Wireless Use
VLF	3 – 30 kHz	Navigation, submarine comm.
LF	30 – 300 kHz	AM broadcasting (long wave)
MF	300 kHz – 3 MHz	AM radio broadcasting
HF	3 – 30 MHz	Shortwave radio
VHF	30 – 300 MHz	FM radio, TV broadcast
UHF	300 MHz – 3 GHz	Cellular mobile, Wi-Fi, Bluetooth
SHF	3 – 30 GHz	Satellite links, 5G mmWave, radar
EHF	30 – 300 GHz	Advanced radar, backhaul links

Wireless communication systems mainly operate in the UHF and SHF bands – they balance range, antenna size, and data capacity for mobile and satellite links.

Why Frequency Spectrum Allocation Is Needed

- **Definition:** spectrum allocation is the process of assigning specific frequency bands to particular radio services or users, to prevent interference and ensure orderly, efficient use of the spectrum.
- The frequency spectrum is a scarce, finite natural resource — it cannot be manufactured or increased, only shared and reused.
- Without allocation, multiple services transmitting on the same frequency would interfere with one another, making communication unusable.

Spectrum management happens at two levels:



Licensed vs Unlicensed Bands

Licensed Bands

- Allocated by regulator (ITU, FCC, TRAI/DoT) for exclusive use
- Assigned to specific operators via auction/allotment
- Example: AMPS cellular got 25 MHz/direction in 800–900 MHz (FCC, 1981)
- Modern cellular mobile: ~700–2600 MHz (2G/3G/4G/5G)
- Interference-protected → guaranteed quality of service

Unlicensed Bands

- Open for public use under regulatory power limits
- No individual license required (ISM bands)
- Wi-Fi and Bluetooth: 2.4 GHz and 5 GHz ISM bands
- Shared simultaneously among many devices/users
- More prone to congestion and interference

Section 1.3

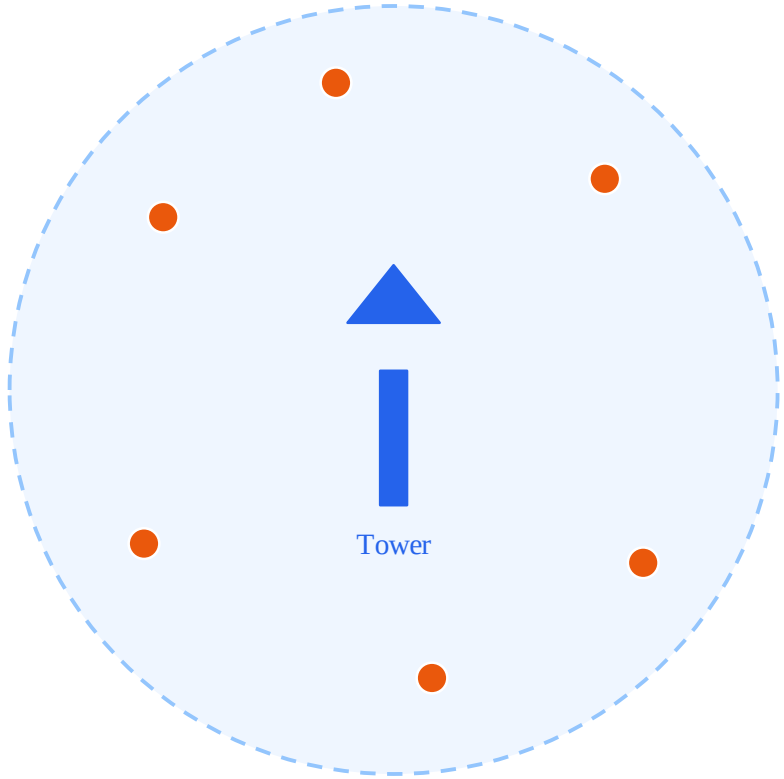
Introduction to Cellular Communication

Before Cellular Systems: One Big Transmitter

- Early mobile radio systems used a single, very powerful transmitter on a tall tower to cover an entire city.

Problem:

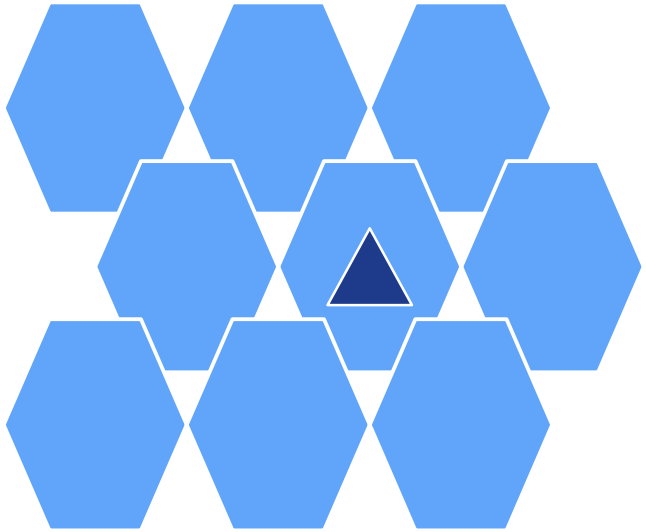
- Only a small, fixed number of channels were available in total
- Since one transmitter served the whole city, a channel used by one caller could not be reused anywhere else nearby
- Result: only a handful of people could talk at the same time, no matter how big the city was
- **Simply increasing transmit power only increases coverage — it does NOT increase the number of users that can be served.**
- *History: proposed by Bell Labs in 1947, detailed in a 1971 FCC proposal, first deployed (AMPS) in Chicago, 1983.*



Whole city — very few channels to share among all users

The Cellular Solution

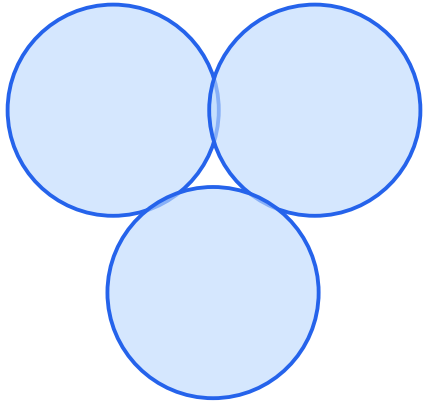
- Idea: instead of one powerful tower, split the city into many small regions called cells.
- **Definition (Cell):** the small geographic area covered by one low-power base station (BS).
- Because each base station transmits at low power, its signal does not travel far — so the same channel can be reused in another cell that is far enough away, without interference.
- **This reuse of channels across many small cells is what lets a cellular network serve a very large number of users with a limited amount of spectrum — hence the name "cellular".**



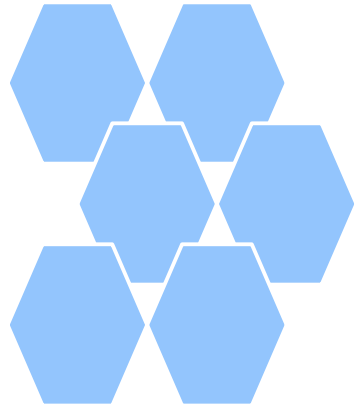
Each small cell has its own low-power base station; far-apart cells can reuse the same channels

Why Are Cells Drawn as Hexagons?

- A base station's antenna actually radiates in all directions, so its real coverage area is a circle.
- Problem: circles cannot cover a map without either leaving gaps or overlapping each other.
- A hexagon is the shape that is closest to a circle, and it can tile a flat area with no gaps and no overlaps.
- **So engineers use hexagons on paper to plan cellular networks — the real coverage area is never a perfect hexagon.**



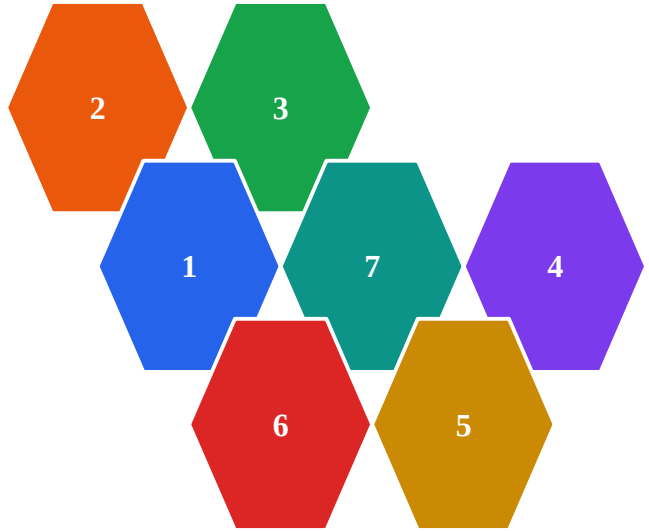
Circles: gaps or overlaps



Hexagons: no gaps, no overlaps

Frequency Reuse

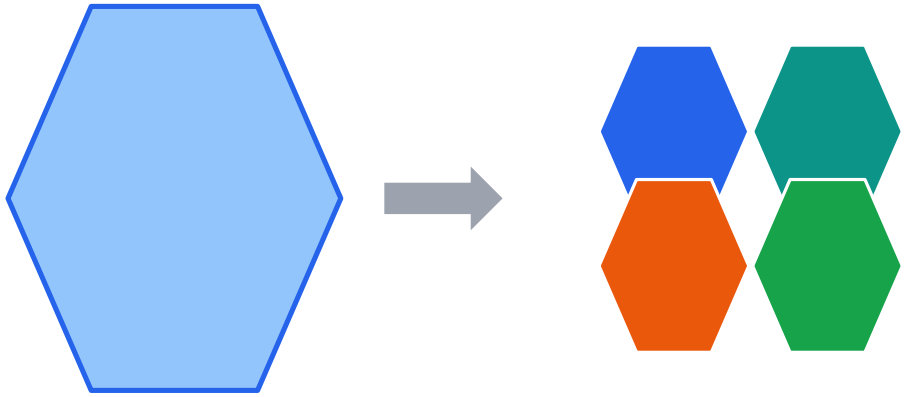
- **Definition:** Frequency reuse is the technique of using the same set of radio frequencies in different cells that are spaced far enough apart to avoid interference.
- Cluster: a group of N adjacent cells that together use the entire set of available channels exactly once.
- Frequency reuse factor = 1/N — each cell uses only a fraction of total channels.
- Only certain cluster sizes are geometrically valid: $N = i^2 + ij + j^2$ ($i, j \geq 0$). Common values: N = 4, 7, 12.
- **Co-channel reuse ratio: $Q = D/R = \sqrt{(3N)}$**
- **System capacity: $C = M \times k \times N$ (M = clusters, k = channels/cell)**



A 7-cell cluster (N = 7): each numbered cell uses a distinct frequency set; the pattern repeats across the network.

Cell Splitting

- Technique used to increase capacity in areas of high traffic density.
- A congested cell is subdivided into smaller cells, each with its own base station and lower transmit power.
- Smaller cells allow frequencies to be reused more often within the same geographic area → higher capacity.
- Trade-off: more base stations to install and more frequent handoffs.

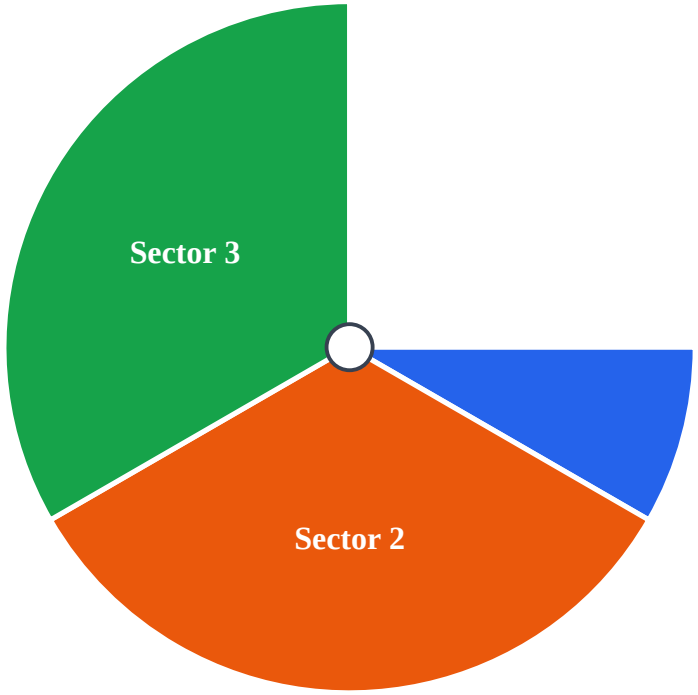


1 macro cell

4 smaller cells

Sectoring

- Technique used to reduce co-channel interference without reducing cluster size.
- A cell's omnidirectional antenna is replaced by directional antennas, each covering a sector — commonly 3 sectors of 120°.
- Each sector only “sees” a fraction of the surrounding co-channel cells, lowering interference.
- Improves signal-to-interference ratio and capacity without splitting cells.



120° directional antennas at the base station

Summary

Today we learned:

- Need for wireless communication
- Elements of a communication system
- Wireless transmission and path loss (qualitative)
- ITU electromagnetic spectrum classification
- Licensed vs unlicensed frequency bands
- Cellular concept and hexagonal cell geometry
- Frequency reuse (cluster size, reuse ratio $Q = \sqrt{3N}$)
- Cell splitting and sectoring as capacity techniques